Practice 2 E02
Charity Donor Contact Masking

Submission Template

Level	Duration	Marks	Tasks	Screenshots
Easy	15 minutes	10	4	4

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Catalog: cat_ltm10877547 Date: _____

Notebook worksheet pipeline or dashboard location: _____

Business scenario

A charity needs donor reporting without exposing complete phone numbers. Protect the phone column of a governed table while preserving missing contacts and operational attributes.

Environment and object names

Use your existing assigned catalog and the fixed schema practice_2. Create that schema if it does not exist. Create only this question's new external volume at its assigned ADLS child path. Use the instructor-provided full URI, not an external-location root.

Item	Required value
Catalog	Your existing assigned catalog
Schema	practice_2
External volume	p2_e02_files
Relative ADLS path	practice_2/e02/files/ beneath your assigned trainee root
Input folder	/Volumes/<trainee_catalog>/practice_2/p2_e02_files/input/
Required object	Managed Delta table p2_e02_donors
Required object	Column mask rule p2_e02_phone_mask

The instructor provides a unique trainee root and complete question URI before timing. The ten question paths are separate. Volumes store source files; managed Delta tables and pipeline outputs use schema managed storage outside volume paths.

Keep supplied file and folder spelling exact. Select your catalog in each execution context, including pipeline destinations and dashboard datasets. Questions are independent; original implementations and required saved outputs must remain reviewable.

Dataset description

P2_E02_Charity_Donor_Contacts.csv

CSV with a header row; 8 data rows and 7 columns. Blank cells mean missing values. Dates use YYYY-MM-DD; Boolean values use true or false.

Column	Required type	Description
donor_id	Text	Donor id
donor_name	Text	Donor name
phone_number	Text	Ten-character synthetic contact; two missing values
city	Text	City
donor_tier	Text	Donor tier
is_active	Boolean true false	Is active
joined_date	Date YYYY-MM-DD	Joined date

Business rules and required columns

1. Load original values into the managed table. Attach an enforced column mask to phone_number; do not permanently replace stored phone values or create only a masked report.

2. Every nonmissing phone has ten characters. Display exactly six asterisks followed by its last four characters. Preserve missing values. Apply this behavior to every querying user without identity exemptions.

3. Retain all eight rows. Other columns remain unchanged. All donor names and contacts are synthetic.

Implementation flow

Prepare access → process data → compare expected outputs → capture evidence.

Task 1 Prepare governed source access

3 minutes | 2 marks | Exactly 1 screenshot

1. Use your existing assigned catalog. Create practice_2 if it does not exist and select it. Do not create a catalog.

2. Create a new EXTERNAL volume named p2_e02_files at the instructor-assigned full cloud URI for this question under the approved ADLS external location. Verify external type and the assigned storage location. Do not create a storage credential or external location.

3. Create an input folder in this volume and upload P2_E02_Charity_Donor_Contacts.csv. Preserve the exact filenames.

Expected output

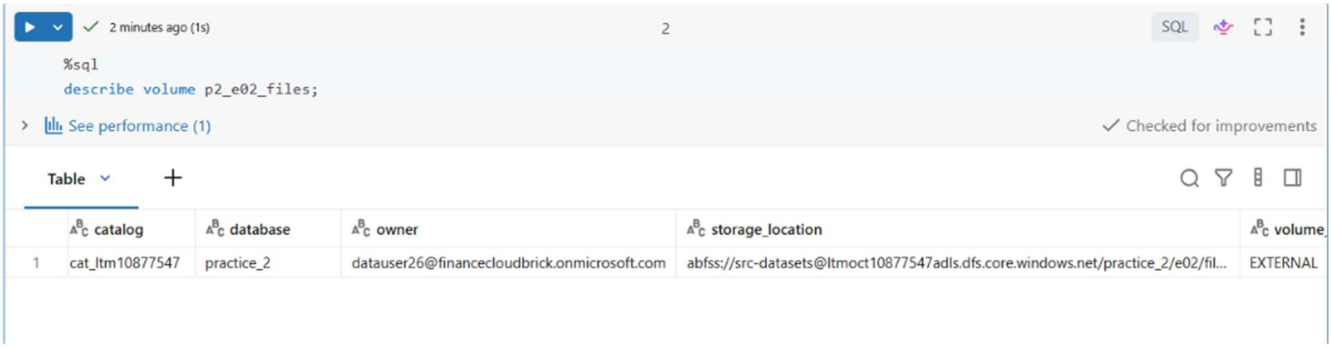
The question external volume, assigned URI and required input file or files are visible.

Required screenshot

External-volume details showing your catalog, practice_2, external type and assigned URI, with every required input filename visible.

Filename: P2_E02_T01.png. Place it in Task 1 of the matching submission template. Your implementation and output must be readable together where requested. Related panels may be captured in one image; tiny text or unreadable stitched images are not acceptable.

Paste the one required screenshot below



Evidence reference or workspace location:
/Workspace/Users/datauser26@financecloudbrick.onmicrosoft.com/newseteasy2

Task 2 Create the original donor table

3 minutes | 2 marks | Exactly 1 screenshot

1. Read with declared types and load original values into p2_e02_donors as managed Delta.

2. Record baseline row count and missing phone count before applying protection.

Expected output

8 rows; 2 missing phones DN203 and DN207; 6 nonmissing phones.

Required screenshot

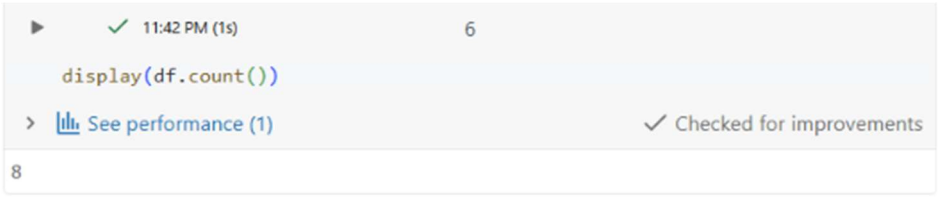
Your baseline loading implementation, table identity and row and missing-phone counts. Plain phone numbers need not be shown.

Filename: P2_E02_T02.png. Place it in Task 2 of the matching submission template. Your implementation and output must be readable together where requested. Related panels may be captured in one image; tiny text or unreadable stitched images are not acceptable.

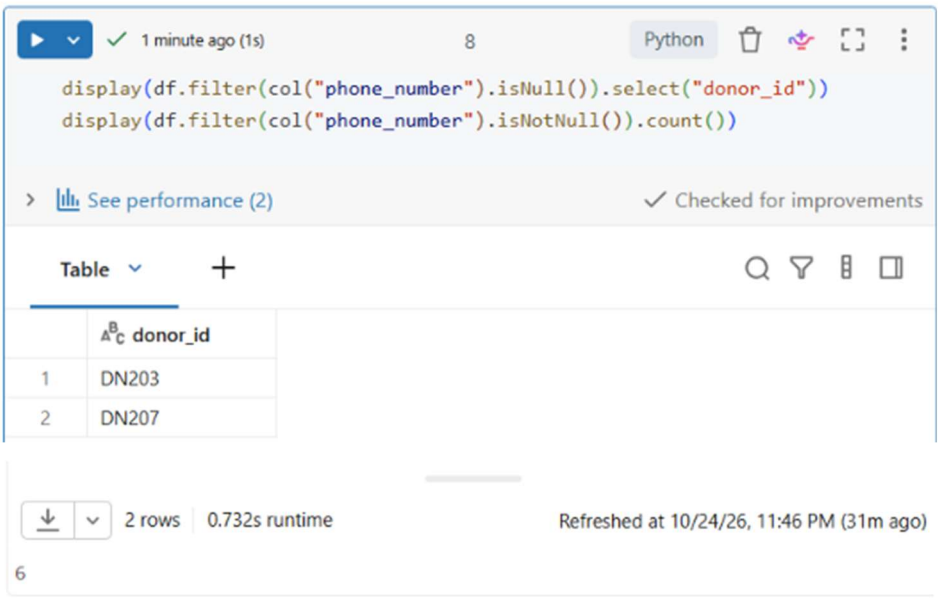
Paste the one required screenshot below











Evidence reference or workspace location:
/Workspace/Users/datauser26@financecloudbrick.onmicrosoft.com/newseteasy2

Task 3 Define and attach contact protection

5 minutes | 3 marks | Exactly 1 screenshot

1. Create the named reusable mask implementing the exact last-four-character rule and missing-value behavior.

2. Attach it to the actual phone_number table column without altering stored contacts.

Expected output

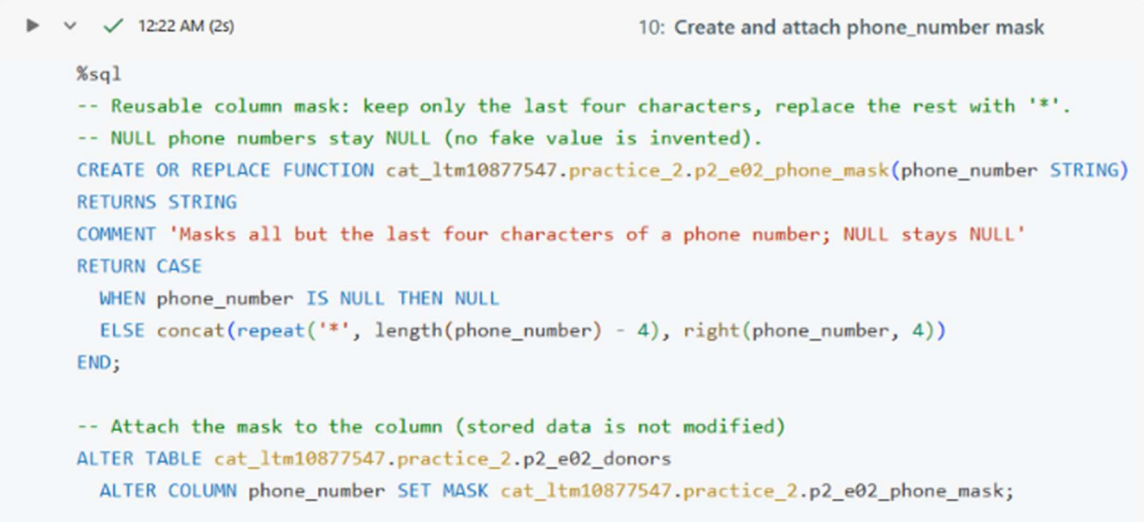
The column has the named enforced mask attached.

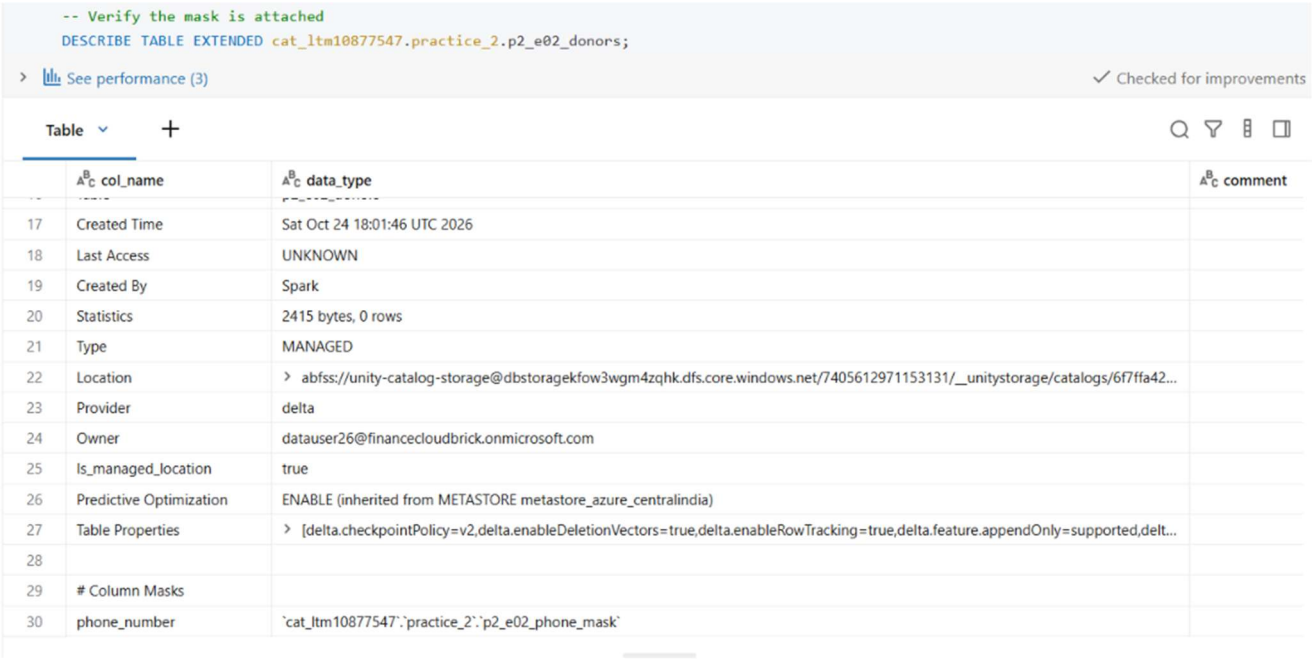
Required screenshot

Your masking implementation and table column metadata identifying the attached mask.

Filename: P2_E02_T03.png. Place it in Task 3 of the matching submission template. Your implementation and output must be readable together where requested. Related panels may be captured in one image; tiny text or unreadable stitched images are not acceptable.

Paste the one required screenshot below





Evidence reference or workspace location:
/Workspace/Users/datauser26@financecloudbrick.onmicrosoft.com/newseteasy2

Task 4 Verify protected donor output

4 minutes | 3 marks | Exactly 1 screenshot

1. Read donor_id, phone_number, city and donor_tier directly from the protected table, ordered by donor_id.

2. Verify six masked phones, two missing values and preserved row count and other attributes.

Expected output

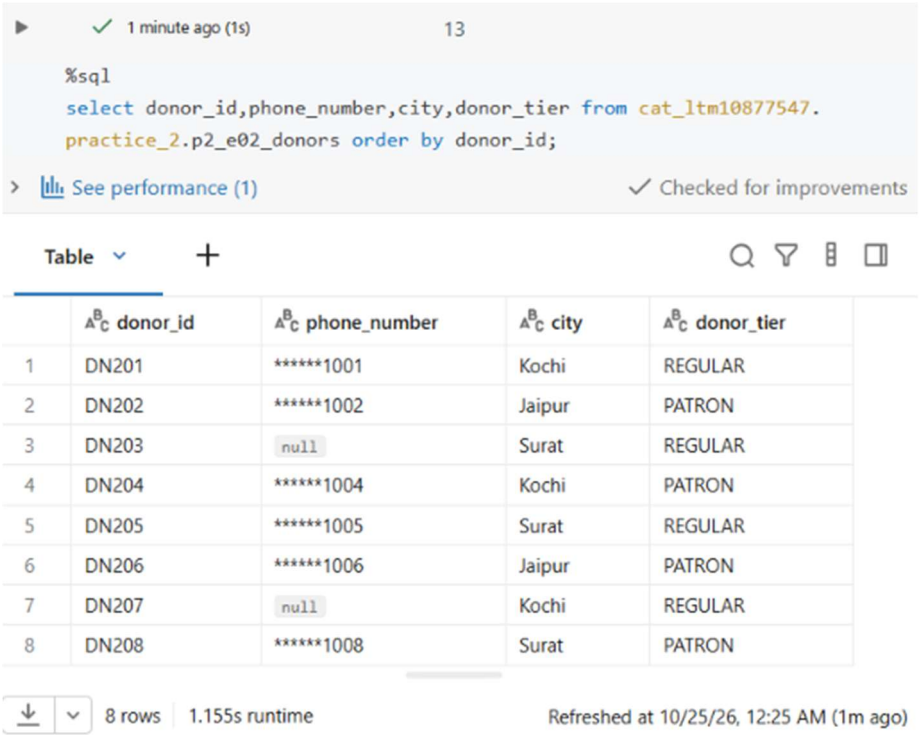
DN201: ******1001 DN202: ******1002 DN203: NULL DN204: ******1004 DN205: ******1005 DN206: ******1006 DN207: NULL DN208: ******1008.

Required screenshot

Your checks and all eight protected output rows with masked and missing counts.

Filename: P2_E02_T04.png. Place it in Task 4 of the matching submission template. Your implementation and output must be readable together where requested. Related panels may be captured in one image; tiny text or unreadable stitched images are not acceptable.

Paste the one required screenshot below



1 minute ago (1s) 13

```
%sql
select donor_id,phone_number,city,donor_tier from cat_ltm10877547.
practice_2.p2_e02_donors order by donor_id;
```

See performance (1) Checked for improvements

Table

	donor_id	phone_number	city	donor_tier
1	DN201	******1001	Kochi	REGULAR
2	DN202	******1002	Jaipur	PATRON
3	DN203	null	Surat	REGULAR
4	DN204	******1004	Kochi	PATRON
5	DN205	******1005	Surat	REGULAR
6	DN206	******1006	Jaipur	PATRON
7	DN207	null	Kochi	REGULAR
8	DN208	******1008	Surat	PATRON

8 rows 1.155s runtime Refreshed at 10/25/26, 12:25 AM (1m ago)

Evidence reference or workspace location:
/Workspace/Users/datauser26@financecloudbrick.onmicrosoft.com/newseteasy2

Expected final output

Donor	Displayed phone	City	Tier
DN201	******1001	Kochi	REGULAR
DN202	******1002	Jaipur	PATRON
DN203	NULL	Surat	REGULAR
DN204	******1004	Kochi	PATRON
DN205	******1005	Surat	REGULAR
DN206	******1006	Jaipur	PATRON
DN207	NULL	Kochi	REGULAR
DN208	******1008	Surat	PATRON

The instructor verifies supported governed compute and grants policy creation and attachment permissions before timing.

Evaluation criteria and screenshot map

Task	Criterion	Marks
1	Correct catalog schema external volume and URI	1
1	Correct input folder and all source files	1
2	Correct governed table with original contact data	1
2	Correct baseline counts	1
3	Correct six-asterisk and last-four behavior with missing preservation	2
3	Mask attached to the actual table column	1
4	Actual table output shows all expected masked and missing values	2
4	Rows and nonmasked attributes preserved	1

Task	Time minutes	Marks	Screenshots	Filename
1	3	2	1	P2_E02_T01.png
2	3	2	1	P2_E02_T02.png
3	5	3	1	P2_E02_T03.png

Task	Time minutes	Marks	Screenshots	Filename
4	4	3	1	P2_E02_T04.png

Total: 15 minutes, 10 marks and exactly 4 screenshots. No extra screenshots are required.

Marks are awarded for the stated implementation and observable task outcomes. A screenshot supports the criterion assigned to its task; it does not create an additional criterion. Retain original work so that implementation and saved outputs remain reviewable.

Submission requirements

Submit the matching template with exactly 4 readable task screenshots. Retain your original implementation and required objects in your trainee workspace folder and add their names or links to the template. Confirm expected outputs and fixed names. Preserve table history and change evidence. No additional report or validator is required.